Abdul-Gafur Saeed

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SUMMARY

Applied AI and Data Systems Specialist with 4+ years of software engineering and digital systems experience, including 2+ years of applied AI/ML research and engineering work spanning clinical deep learning, computer vision, and enterprise generative AI. At the African Union Commission (AUC), architected enterprise RAG workflows (LangGraph, pgvector, E5), implemented dual-runtime sovereign AI (Azure OpenAI and locally hosted Qwen 8B), and operationalized continental monitoring frameworks using DHIS2 and Power BI DAX. Combines model development and rigorous evaluation with the engineering of secure, deployable data systems. Experience includes offline-first EMR decision support for health logistics, ECG arrhythmia classification with neural network pruning, and technical leadership for platforms supporting 35,000+ MSME users across six African countries. Focused on scalable digital public goods that address infrastructure constraints and advance health, development, and equity across Sub-Saharan Africa. Certified Azure and AWS Developer Associate.

EDUCATIONAL BACKGROUND

University of East London – London, United Kingdom **2026 - 2027**

Master of Science, Computer and Information Science (Online)

University of Business and Integrated Development Studies – Wa, Ghana **2021 – 2025**

Bachelor of Science, Computer Science

BSc Research: Deep Learning for Arrhythmia Detection: evaluated CNNs on the MIT-BIH ECG benchmark and applied first-order movement pruning, achieving 3.8× parameter compression while preserving 96.2% sensitivity for ventricular ectopic beats in low-resource point-of-care and wearable diagnostic contexts.

PROFESSIONAL WORK EXPERIENCE

Digital & Innovation Expert (AU Tech Fellow) **April 2026 – Present**

African Union Commission, MIS Application Unit – Addis Ababa, Ethiopia (On-site)

- Architected and developed AskAU, an enterprise RAG knowledge assistant indexing 200+ institutional policy documents, transforming unstructured repositories into an evidence-grounded search resource for AUC personnel.
- Engineered a modular, vendor-agnostic LLM gateway supporting dual-runtime inference (Azure OpenAI for rapid prototyping and locally hosted Qwen 8B on AUC sovereign infrastructure), ensuring data privacy, offline resilience, and zero external token dependencies.
- Built an end-to-end multi-step retrieval pipeline using Python, LangGraph, and FastAPI, integrating SharePoint (Graph API) and developing an open-source hybrid retrieval engine (PostgreSQL/pgvector semantic search + full-text search) that optimized domain-specific policy recall.
- Executed rigorous evaluation protocols on a ground-truth benchmark suite of 50+ validated query-response pairs across 5 institutional treaties, achieving 88% Retrieval Recall (Recall@5), 82% Context Precision, and 94% Citation Accuracy.
- Deployed the containerized application with progressive streaming in an operational pilot across 20 MISD analysts, incorporating user feedback to refine chunking strategies, retrieval behaviour, and access-control boundaries.
- Implemented enterprise responsible AI guardrails, including strict source-grounded citations for auditability, permission-aware access filtering, and low-resource institutional language testing to mitigate linguistic bias and unsupported generation.
- Co-designed the data architecture for the African Blue Economy Dashboard, operationalizing 30 KPIs and the AUBEI scoring framework using DHIS2 and Power BI DAX to standardize multi-sector reporting and benchmark performance across AU Member States, enabling evidence-based maritime policy and resource allocation decisions.
- Engineered interoperable country-level data collection workflows using KoboToolbox XLSForms, DHIS2 validation rules, and Python/PostgreSQL ETL pipelines, standardizing national datasets into verified indicators adaptable to health-logistics reporting.
- Developed backend APIs and automated data services for the Gender Observatory Hub (.NET, Next.js, React, PostgreSQL, Power BI REST API), operationalizing 30 statutory gender indicators across AU Member States and Regional Economic Communities (RECs), reducing continental reporting cycles and providing member states with verified gender equity benchmarks for policy planning.
- Contributed to the AUC 6-Year ICT Strategy by evaluating institutional gaps in AI adoption and software development; authored the Application Development Lifecycle SOP, Application Security Guidelines (OWASP ASVS and ISO/IEC 27001-aligned), Technology Selection Guidelines, and Software Acquisition Standards to enforce strict institutional compliance across all directorate applications.
- Advanced institutional AI literacy by authoring 3 practical implementation playbooks, delivering hands-on GenAI workshops for 60+ technical and non-technical staff across 3 directorates, and mentoring engineering interns.

Chief Technology Officer **March 2022 - March 2026**

Paahibu Space – Wa, Ghana (Hybrid)

- Directed technical strategy, architecture, and engineering delivery of Ndiara, a multi-country platform supporting approximately 35,000 users across six African countries, enabling women-led MSMEs through integrated digital business and financial services.
- Led a 5-member multidisciplinary engineering team spanning product management, frontend/backend engineering, QA, and cloud infrastructure, directing technical planning, system design, and delivery coordination.
- Architected an integrated digital ecosystem spanning e-commerce, learning and development, HR management, point of sale, accounting, and digital wallet services, consolidating essential business operations into a unified platform.
- Engineered REST API middleware and integrated six external platforms, enabling interoperability across business services and streamlining fragmented operational workflows in multi-country contexts.
- Automated program reporting workflows, reducing turnaround from 3–5 days to under 10 minutes, and achieved a 74% 90-day user retention rate.

Founder and Lead Architect **July 2022 – Present**

SmartSakuu – Ghana (Remote)

- Founded and engineered SmartSakuu, an AI-powered education and school management platform connecting administration, curriculum delivery, and personalized learning for African classrooms.
- Built an AI tutor aligned with Ghana Education Service (GES) and NaCCA standards for BECE and WASSCE preparation, analyzing student quiz results to pinpoint learning gaps and automatically generate personalized practice questions.
- Enforced child data protection and curriculum fidelity by implementing automated PII scrubbing on student records, dialect/accent-tolerant prompt templates, and mandatory human-in-the-loop teacher validation for generated instructional materials.
- Integrated school workflows by extending InfixEdu (admissions, records, fee tracking), partnering with GNAPS Wa Municipal to pilot the platform across 8 partner schools reaching 600+ students and teachers.

DevSecOps Engineer **July 2025 - February 2026**

CloudFruition – London, UK (Remote)

- Engineered cloud application infrastructure across AWS and Microsoft Azure, containerizing microservices with Docker and Docker Compose to improve deployment portability and local testing fidelity.
- Developed .NET-based integration services for CloudSnap, enabling automated cross-cloud backup, disaster recovery, and workload migration across 15+ microservices between AWS and Azure, reducing recovery time objectives (RTO) by 40%.
- Implemented and maintained GitHub Actions CI/CD pipelines, automating build verification, automated testing, and zero-downtime release workflows across 20+ deployment pipelines, cutting deployment lead times from hours to under 15 minutes.
- Hardened cloud and pipeline security by enforcing zero-trust access across 50+ IAM roles, protecting 100+ production API secrets via AWS Secrets Manager, and maintaining 99.9% service uptime via CloudWatch monitoring.

SELECTED AI & HEALTH ENGINEERING PROJECTS

Offline-First Point-of-Care EMR & Health Logistics Decision Support **2024 – 2025**

Lead Architect & Developer – Health Informatics Prototype for Low-Connectivity Facilities

- Architected an offline-first Electronic Medical Record (EMR) and dispensary inventory prototype for rural clinics in Wa, Upper West Region of Ghana, combining local SQLite caching with asynchronous background REST synchronization to sustain point-of-care clinical workflows during network outages.
- Implemented a pharmaceutical decision-support engine applying Average Monthly Consumption (AMC) velocity and First-Expired, First-Out (FEFO) batch tracking to estimate Days of Stock Remaining, flag potential stockout risk, and identify inter-facility redistribution before medicine expiration.
- Validated inventory and replenishment algorithms against retrospective primary clinic dispensing logs (evaluating stockout alert precision against historical stock runs), tracking revolving drug-fund cash releases against facility budget ceilings to diagnose procurement bottlenecks.

Clinical ECG Arrhythmia Classification & Movement Pruning **2024 – 2025**

Academic Research (BSc Thesis) – AI in Healthcare & Low-Resource Diagnostics

- Trained a deep Convolutional Neural Network (CNN) on multi-lead ECG recordings from the benchmark MIT-BIH Arrhythmia Database to classify cardiac arrhythmias for resource-constrained clinical settings.
- Implemented first-order movement pruning using an inter-patient train/test split, achieving a 3.8× parameter compression and substantial FLOP reduction while preserving 96.2% sensitivity for ventricular ectopic beats for wearable and edge deployment.

ScrapTrace – Computer Vision E-Waste Classification **2026**

GIZ Data Lab E-Waste Hackathon – Team project, winning entry

- Curated and preprocessed an image dataset of 1,232 e-waste items and fine-tuned a MobileNetV2 CNN using TensorFlow, Keras, and OpenCV; evaluated on a stratified 248-image held-out test split, achieving 87.10% accuracy (216/248) and a 0.8135 macro F1 score across 5 e-waste categories (e.g., >91% precision on high-contrast PCBs).
- Co-architected TypeScript/Zod shared service contracts governing image-inference validation, evidence integrity, lifecycle-state transitions, and idempotency across web and vision microservices.

SKILLS

AI, Machine Learning & Generative AI: Large Language Models (LLMs), RAG, LangChain, LangGraph, prompt engineering, hybrid retrieval, E5 embeddings, local inference (Qwen 8B), Azure OpenAI, Vector Databases (pgvector), Convolutional Neural Networks (CNNs), MobileNetV2, TensorFlow, Keras, OpenCV, model pruning and compression (movement pruning), model evaluation (Recall@k, Context Precision, Citation Accuracy, Macro F1, sensitivity), inventory and consumption modelling (AMC, Min-Max, DSR).

Data Systems & Health Informatics: DHIS2, KoboToolbox (XLSForms), EMR architecture, offline-first synchronization, Power BI (DAX, REST APIs), PostgreSQL, SQL optimization, ETL pipelines, comparative analytics, indicator and scoring frameworks, health-logistics decision support.

Cloud, DevSecOps & Architecture: Microsoft Azure, AWS, Docker, Docker Compose, GitHub Actions, CI/CD, enterprise solution architecture, REST APIs, microservices, IAM least privilege, secrets management.

Software Engineering & Responsible AI Controls: Python (FastAPI, Django), .NET, Node.js, Next.js, React, TypeScript, OWASP ASVS 4.0, ISO/IEC 27001-aligned controls, PII scrubbing, permission-aware retrieval, source-grounded citations, linguistic bias mitigation, human-in-the-loop oversight, technical playbooks.

CERTIFICATIONS AND AWARDS

- Microsoft Certified: Azure Developer Associate
- AWS Certified Developer – Associate
- AWS Certified Cloud Practitioner
- Postman API Student Expert
- Winner (team) – GIZ Data Lab E-Waste Hackathon 2026
- NoniHub Award – AI Innovation
- Finalist – MTN Ayoba Hackathon 2021

LANGUAGES

- Waale – Native
- English – Fluent / Expert
- French – Beginner
- Arabic – Beginner